

exam1

March 30, 2025

```
[10]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from sklearn import linear_model
from itertools import combinations
from sklearn.metrics import mean_squared_error

df = pd.read_csv("prostate.csv")

train = df[df["train"] == "T"]
test = df[df["train"] == "F"]

x_train = train[["lcavol", "lweight", "age", "lbph", "svi", "lcp", "gleason",
    ↪ "pgg45"]]
y_train = train["lpsa"]
x_test = test[["lcavol", "lweight", "age", "lbph", "svi", "lcp", "gleason",
    ↪ "pgg45"]]
y_test = test["lpsa"]
```

3a) Standard Linear Regression Model using OLS

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[11]: linear_regression = linear_model.LinearRegression()
linear_regression.fit(x_train, y_train)

lr = sm.OLS(y_train, x_train)
results = lr.fit()
print(results.summary())

# R^2
print("R^2: ", results.rsquared)

# P-values
print("P-values: ", results.pvalues)

# Significant predictors using 0.05 significance level
significant_predictors = results.pvalues[results.pvalues < 0.05]
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print("Significant predictors: ", significant_predictors)

# Training and Testing Errors
train_predictions = linear_regression.predict(x_train)
test_predictions = linear_regression.predict(x_test)
train_error = np.mean((y_train - train_predictions) ** 2)
test_error = np.mean((y_test - test_predictions) ** 2)
print("Training Error: ", train_error)
print("Testing Error: ", test_error)

```

OLS Regression Results

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=====
Dep. Variable:          lpsa      R-squared (uncentered):
0.941
Model:                  OLS      Adj. R-squared (uncentered):
0.933
Method:                 Least Squares      F-statistic:
117.6
Date:                   Sat, 29 Mar 2025      Prob (F-statistic):
2.43e-33
Time:                   22:12:28      Log-Likelihood:
-67.549
No. Observations:      67      AIC:
151.1
Df Residuals:          59      BIC:
168.7
Df Model:               8
Covariance Type:       nonrobust
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```

	coef	std err	t	P> t	[0.025	0.975]
lcavol	0.5706	0.104	5.463	0.000	0.362	0.780
lweight	0.6461	0.189	3.417	0.001	0.268	1.024
age	-0.0181	0.013	-1.380	0.173	-0.044	0.008
lbph	0.1383	0.066	2.100	0.040	0.007	0.270
svi	0.7414	0.296	2.506	0.015	0.149	1.333
lcp	-0.2068	0.110	-1.887	0.064	-0.426	0.013
gleason	0.0120	0.133	0.090	0.928	-0.254	0.278
pgg45	0.0087	0.005	1.844	0.070	-0.001	0.018

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Omnibus:                0.837      Durbin-Watson:                1.710
Prob(Omnibus):          0.658      Jarque-Bera (JB):              0.382
Skew:                   -0.156      Prob(JB):                      0.826
Kurtosis:               3.198      Cond. No.                      252.
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```

Notes:

[1] R^2 is computed without centering (uncentered) since the model does not contain a constant.

[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

R^2 : 0.9409776293201828

P-values: lcavol 9.864471e-07

lweight 1.152215e-03

age 1.726477e-01

lbph 4.001430e-02

svi 1.498475e-02

lcp 6.414204e-02

gleason 9.284580e-01

pgg45 7.020806e-02

dtype: float64

Significant predictors: lcavol 9.864471e-07

lweight 1.152215e-03

lbph 4.001430e-02

svi 1.498475e-02

dtype: float64

Training Error: 0.4391997680583344

Testing Error: 0.5212740055076022

3b) AIC Value vs Degree of Freedom graph

```
[ ]: def AIC(train_data, train_label):
    lm = linear_model.LinearRegression().fit(train_data, train_label)
    predict_label = lm.predict(train_data)
    degree_freedom = train_data.shape[1]
    sample_size = train_data.shape[0]
    SSE = np.dot((train_label - predict_label), (train_label - predict_label))
    return sample_size * np.log(SSE / sample_size) + 2 * degree_freedom

feature_names = list(x_train.columns.values)
min_AIC = []
min_AIC.append(y_train.var())
plt.scatter(1, y_train.var()) # when there is only the intercept term, plot variance
for i in range(1, 8):
    names = list(
        combinations(feature_names, i)
    ) # generate the combination of i feature names and convert it to a list
    print(names)
    aic_list = [] # define a list to store AIC values
    for each_element in names:
        aic_list.append(
            AIC(x_train[list(each_element)], y_train)
        ) # append BIC values to the list
```

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min_AIC.append(np.min(aic_list))
min_index = np.argmin(aic_list)
print(aic_list)
for each_value in aic_list:
    plt.scatter(x=i + 1, y=each_value) # plot AIC values

plt.plot([1, 2, 3, 4, 5, 6, 7, 8], min_AIC)
plt.title("AIC")
plt.xlabel("Degree of Freedom")
plt.ylabel("AIC Value")
plt.show()

print("The minimum AIC values are: ", min_AIC)
print("Minimum AIC value is: ", np.min(min_AIC))
print(
    "The minimum AIC value is obtained using the corresponding features: ",
    names[min_index],
)

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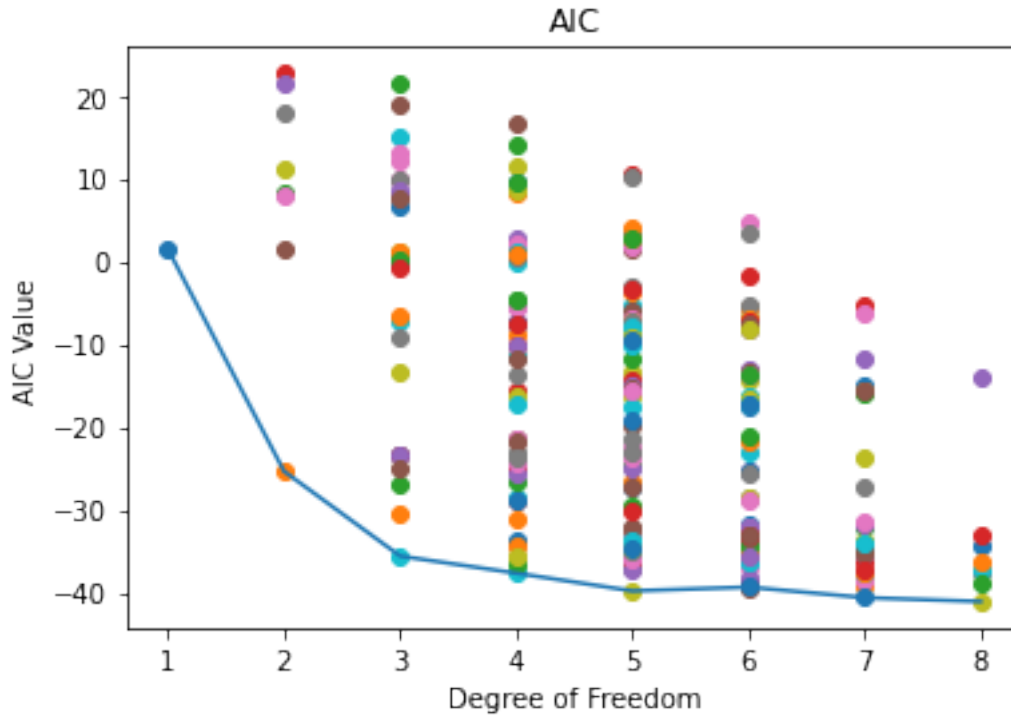
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```

The minimum AIC values are: [1.458809773002296, -25.37360907803762, -35.61679636983291, -37.68291356508498, -39.82507244977625, -39.364852213098764, -40.6393931650699, -41.10281207095607]

Minimum AIC value is: -41.10281207095607

The minimum AIC value is obtained using the corresponding features: ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'lcp', 'pgg45')

3c)

```
[21]: def BIC(train_data, train_label):
    lm = linear_model.LinearRegression().fit(train_data, train_label)
    predict_label = lm.predict(train_data)
    degree_freedom = train_data.shape[1]
    sample_size = train_data.shape[0]
    SSE = np.dot((train_label - predict_label), (train_label - predict_label))
    return (
        sample_size * np.log(SSE / sample_size) + np.log(sample_size) *
        degree_freedom
    )

feature_names = list(x_train.columns.values)
min_BIC = []
min_BIC.append(y_train.var())
```

```

plt.scatter(1, y_train.var()) # when there is only the intercept term, plot
    ↪ variance
for i in range(1, 8):
    names = list(
        combinations(feature_names, i)
    ) # generate the combination of i feature names and convert it to a list
    print(names)
    bic_list = [] # define a list to store BIC values
    for each_element in names:
        bic_list.append(
            BIC(x_train[list(each_element)], y_train)
        ) # append BIC values to the list
    print(bic_list)
    min_BIC.append(np.min(bic_list))
    for each_value in bic_list:
        plt.scatter(x=i + 1, y=each_value) # plot BIC values
plt.plot([1, 2, 3, 4, 5, 6, 7, 8], min_BIC)
plt.title("BIC")
plt.xlabel("Degree of Freedom")
plt.ylabel("BIC Value")
plt.show()

print("The minimum BIC values are: ", min_BIC)
print("Minimum BIC value is: ", np.min(min_BIC))
print(
    "The minimum BIC value is obtained using the corresponding features: ",
    names[min_index],
)

```

```

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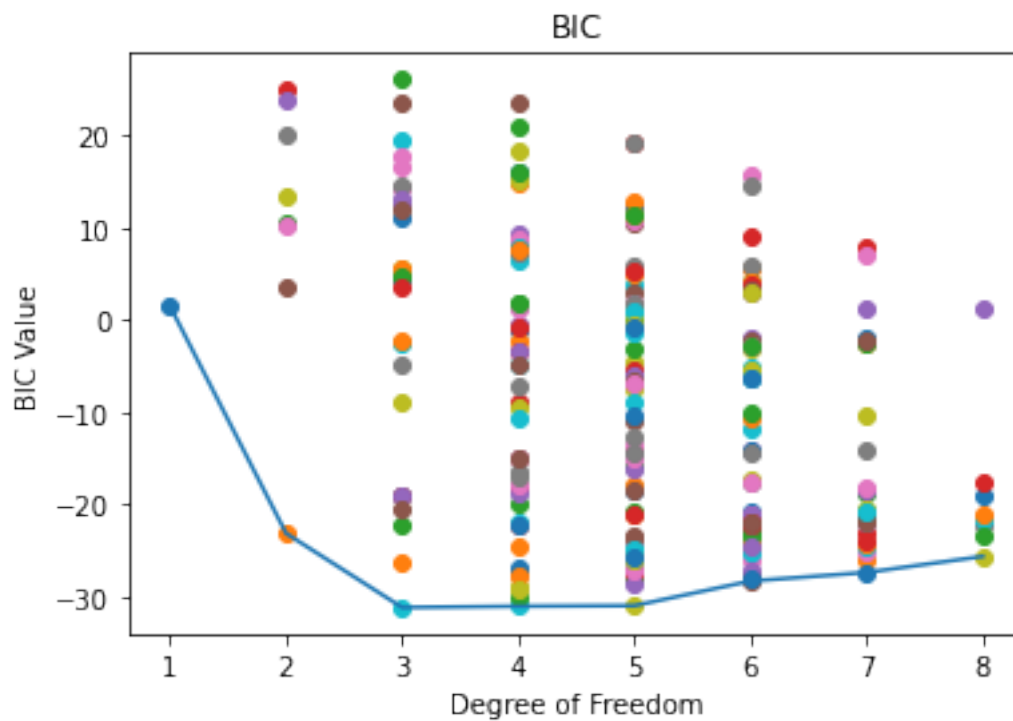
2.889733092533149]

[('lcavol', 'lweight', 'age', 'lbph', 'svi', 'lcp'), ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'gleason'), ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'pgg45'), ('lcavol', 'lweight', 'age', 'lbph', 'lcp', 'gleason'), ('lcavol', 'lweight', 'age', 'lbph', 'lcp', 'pgg45'), ('lcavol', 'lweight', 'age', 'lbph', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'age', 'svi', 'lcp', 'gleason'), ('lcavol', 'lweight', 'age', 'svi', 'lcp', 'pgg45'), ('lcavol', 'lweight', 'age', 'svi', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'age', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'lbph', 'svi', 'lcp', 'gleason'), ('lcavol', 'lweight', 'lbph', 'svi', 'lcp', 'pgg45'), ('lcavol', 'lweight', 'lbph', 'svi', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'lbph', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'svi', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'age', 'lbph', 'svi', 'lcp', 'gleason'), ('lcavol', 'age', 'lbph', 'svi', 'lcp', 'pgg45'), ('lcavol', 'age', 'lbph', 'svi', 'gleason', 'pgg45'), ('lcavol', 'age', 'lbph', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'age', 'svi', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45'), ('lweight', 'age', 'lbph', 'svi', 'lcp', 'gleason'), ('lweight', 'age', 'lbph', 'svi', 'lcp', 'pgg45'), ('lweight', 'age', 'lbph', 'svi', 'gleason', 'pgg45'), ('lweight', 'age', 'lbph', 'lcp', 'gleason', 'pgg45'), ('lweight', 'age', 'svi', 'lcp', 'gleason', 'pgg45'), ('lweight', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45'), ('age', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45')]

[-24.674980668283176, -24.69080702624747, -25.987698379533867, -18.87540554997264, -23.029618496923838, -21.93145832967395, -22.044278363044235, -25.182607234089247, -20.927746687134643, -20.479585094096628, -24.461696360933182, -27.4112374487241, -24.276133984926915, -21.145775666605836, -23.952064184871624, -18.540972033847027, -21.328627526158087, -18.25578986887904, -14.121804356774632, -10.366340306026672, -20.942516654813616, -1.8521214212023729, -2.4530189708741297, -2.6768716655325093, 8.02363664362969, 1.3566299116274827, -2.373226904009236, 7.093874052053469]

[('lcavol', 'lweight', 'age', 'lbph', 'svi', 'lcp', 'gleason'), ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'lcp', 'pgg45'), ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'age', 'lbph', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'age', 'svi', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'lweight', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45'), ('lcavol', 'age', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45'), ('lweight', 'age', 'lbph', 'svi', 'lcp', 'gleason', 'pgg45')]

[-22.293886883777002, -25.66996373521931, -21.784952218560992, -18.99762707010701, -20.982169549722666, -23.480872996734508, -17.47879710235237, 1.3148896914720716]



The minimum BIC values are: [1.458809773002296, -23.168916458646656, -31.207411131050982, -31.068835706912083, -31.006301972212388, -28.341389116143937, -27.4112374487241, -25.66996373521931]

Minimum BIC value is: -31.207411131050982

The minimum BIC value is obtained using the corresponding features: ('lcavol', 'lweight', 'age', 'lbph', 'svi', 'lcp', 'pgg45')